Chapter 9. Epilogue—Life Outside $\mathbb{R}^n$

Differentiable manifolds and Riemannian manifolds

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Overview

We have dealt with submanifolds of euclidean space and forms on open sets of euclidean space: conceptually simple, but important ideas are sometimes obscured by the familiar surroundings, and in mathematics and physics manifolds often occur as abstract spaces. This chapter sketches how to move outside $\mathbb{R}^n$.

§41. Differentiable Manifolds and Riemannian Manifolds

Motivation. So far manifolds sat inside $\mathbb{R}^n$. Abstract manifolds remove the ambient space and keep only coordinate charts. Everything proved for submanifolds, including forms and Stokes' theorem, carries over once tangent vectors are defined intrinsically.

41.1 Differentiable manifolds

Definition 41.1 (differentiable manifold)

Let M be a metric space. Suppose there is a collection of homeomorphisms $\alpha_i : U_i \to V_i$, where U_i is open in $\mathbb{H}^k$ or $\mathbb{R}^k$ and V_i is open in M, such that the V_i cover M. (α_i is a **homeomorphism** if it carries U_i onto V_i one-to-one and both α_i, α_i^{-1} are continuous.) Suppose the α_i overlap with class C^∞: the **transition function** $\alpha_i^{-1} \circ \alpha_j$ is of class C^∞ whenever $V_i \cap V_j$ is non-empty. The α_i are **coordinate patches** on M, as is any other homeomorphism $\alpha : U \to V$ (U open in $\mathbb{H}^k$ or $\mathbb{R}^k$, V open in M) that overlaps the α_i with class C^∞. M with this collection of patches is a **differentiable k-manifold** (of class C^∞).

For $k = 1$ we again allow domains open in $\mathbb{L}^1$. If there is a patch $\alpha : U \to V$ about p with U open in $\mathbb{R}^k$, p is an **interior point**; otherwise a **boundary point**; the boundary points form ∂M. If $\alpha : U \to V$ is a patch about p, then $p \in \partial M$ iff U is open in $\mathbb{H}^k$ and $p = \alpha(\mathbf{x})$ for some $\mathbf{x} \in \mathbb{R}^{k-1} \times 0$ (proof as in Lemma 24.2). Throughout, M denotes a differentiable k-manifold.

Definition 41.2 (orientation)

Patches α_0, α_1 on M **overlap positively** if $\det D(\alpha_1^{-1} \circ \alpha_0) > 0$. If M can be covered by patches overlapping positively, M is **orientable**. An **orientation** of M consists of such a covering together with all other patches overlapping these positively; an **oriented manifold** is a manifold with an orientation. If $\{\alpha_i\}$ is an orientation, $\{\alpha_i \circ r\}$ (r the reflection of $\mathbb{R}^k$) is the **opposite** orientation.

If $\partial M \neq \emptyset$, then ∂M is a differentiable $(k-1)$-manifold without boundary, with patches $\alpha \circ b$ (α a patch about $p \in \partial M$, $b(x_1, \ldots, x_{k-1}) = (x_1, \ldots, x_{k-1}, 0)$), as in Theorem 24.3. If α_0, α_1 overlap positively, so do $\alpha_0 \circ b$ and $\alpha_1 \circ b$ (Theorem 34.1). So if M is oriented, ∂M can be oriented by restricting the patches of the orientation of M; for k even this is the **induced orientation** of ∂M, for k odd its opposite is.

Definition 41.3 (C^∞ maps between manifolds)

Let M, N be differentiable manifolds of dimensions k, n, $A \subset M$ and $f : A \to N$. f is **of class** C^∞ if for each $x \in A$ there are a patch $\alpha : U \to V$ on M about x and a patch $\beta : W \to Y$ on N about $y = f(x)$ such that $\beta^{-1} \circ f \circ \alpha$ is of class C^∞ as a map of a subset of $\mathbb{R}^k$ into $\mathbb{R}^n$. (Independent of the patches, since transition functions are C^∞.) A one-to-one

$f : M \to N$ onto N with f and f^{-1} of class C^∞ is a **diffeomorphism**.

41.2 Tangent vectors

Without a surrounding euclidean space, what is a tangent vector? For M in $\mathbb{R}^n$, a tangent vector at p is the velocity $(p; D\gamma(t_0))$ of a C^∞ curve $\gamma : [a, b] \to M$ with $\gamma(t_0) = p$. For abstract M we cannot differentiate γ, but for a patch $\alpha : U \to V$ about p, $\alpha^{-1} \circ \gamma$ maps a subset of $\mathbb{R}^1$ into $\mathbb{R}^k$, so we may think of the "derivative" of γ at t_0 as the function $\mathbf{v}$ assigning to each patch α about p the matrix $\mathbf{v}(\alpha) = D(\alpha^{-1} \circ \gamma)(t_0)$. If α_0, α_1 are two patches about p, the chain rule gives

$$\mathbf{v}(\alpha_1) = Dg(\mathbf{x}_0) \cdot \mathbf{v}(\alpha_0), \qquad g = \alpha_1^{-1} \circ \alpha_0, \quad \mathbf{x}_0 = \alpha_0^{-1}(p).$$

Definition 41.4 (tangent vector, tangent space)

Given $p \in M$, a **tangent vector** to M at p is a function $\mathbf{v}$ assigning to each coordinate patch $\alpha : U \to V$ about p a k by 1 column matrix $\mathbf{v}(\alpha)$, such that for any two patches α_0, α_1 about p,

$$\mathbf{v}(\alpha_1) = Dg(\mathbf{x}_0) \cdot \mathbf{v}(\alpha_0), \tag{$*$}$$

where $g = \alpha_1^{-1} \circ \alpha_0$ and $\mathbf{x}_0 = \alpha_0^{-1}(p)$. The entries of $\mathbf{v}(\alpha)$ are the **components** of $\mathbf{v}$ with respect to α.

By $(*)$, $\mathbf{v}$ is determined by its components in a single patch, and we may define $(a\mathbf{v} + b\mathbf{w})(\alpha) = a\mathbf{v}(\alpha) + b\mathbf{w}(\alpha)$. The tangent vectors at p form the **tangent space** $\mathcal{T}_p(M)$, a k-dimensional vector space: if $\alpha(\mathbf{x}) = p$, the map $\mathbf{v} \mapsto (\mathbf{x}; \mathbf{v}(\alpha))$ is a linear isomorphism of $\mathcal{T}_p(M)$ onto $\mathcal{T}_{\mathbf{x}}(\mathbb{R}^k)$, whose inverse is denoted $\alpha_* : \mathcal{T}_{\mathbf{x}}(\mathbb{R}^k) \to \mathcal{T}_p(M)$; so $\alpha_*(\mathbf{x}; \mathbf{v}(\alpha)) = \mathbf{v}$. For a C^∞ curve $\gamma : [a, b] \to M$ with $\gamma(t_0) = p$, the **velocity vector** is defined by $\mathbf{v}(\alpha) = D(\alpha^{-1} \circ \gamma)(t_0)$; every tangent vector at p is the velocity vector of some curve.

Tangent vectors as derivations

A common alternative: to $\mathbf{v} \in \mathcal{T}_p(M)$ associate an operator $X_{\mathbf{v}}$ on real-valued C^∞ functions near p, the **derivative with respect to** $\mathbf{v}$. If $\mathbf{v}$ is the velocity of γ at t_0 and $\alpha(\mathbf{x}) = p$, then writing $f \circ \gamma = (f \circ \alpha) \circ (\alpha^{-1} \circ \gamma)$,

$$\frac{d(f \circ \gamma)}{dt}(t_0) = D(f \circ \alpha)(\mathbf{x}) \cdot D(\alpha^{-1} \circ \gamma)(t_0) = D(f \circ \alpha)(\mathbf{x}) \cdot \mathbf{v}(\alpha),$$

which depends only on f and $\mathbf{v}$. So define $X_{\mathbf{v}}(f) = D(f \circ \alpha)(\mathbf{x}) \cdot \mathbf{v}(\alpha)$; it is independent of α, $X_{\mathbf{v}+\mathbf{w}} = X_{\mathbf{v}} + X_{\mathbf{w}}$, $X_{c\mathbf{v}} = cX_{\mathbf{v}}$, and for $M = \mathbb{R}^k$ it is the directional derivative of f with respect to $\mathbf{v}$. It satisfies (1) *locality*: if f, g agree near p, $X_{\mathbf{v}}(f) = X_{\mathbf{v}}(g)$;
(2) *linearity*: $X_{\mathbf{v}}(af + bg) = aX_{\mathbf{v}}(f) + bX_{\mathbf{v}}(g)$;
(3) the *product rule*: $X_{\mathbf{v}}(f \cdot g) = X_{\mathbf{v}}(f)g(p) + f(p)X_{\mathbf{v}}(g)$. These characterize $X_{\mathbf{v}}$: every operator X on C^∞ functions near p satisfying (1)–(3) equals $X_{\mathbf{v}}$ for a unique $\mathbf{v} \in \mathcal{T}_p(M)$. So one can define a tangent vector at p to be such an operator; many authors prefer this "intrinsic" definition, which does not involve coordinate patches explicitly.

41.3 Forms on M

Definition 41.5 (forms on a manifold)

An l-**form** on M is a function ω assigning to each $p \in M$ an alternating l-tensor $\omega(p) \in \mathcal{A}^l(\mathcal{T}_p(M))$. If $\alpha : U \to V$ is a patch about p with $\alpha(\mathbf{x}) = p$, let $T = \alpha_* : \mathcal{T}_{\mathbf{x}}(\mathbb{R}^k) \to \mathcal{T}_p(M)$ and $T^* : \mathcal{A}^l(\mathcal{T}_p(M)) \to \mathcal{A}^l(\mathcal{T}_{\mathbf{x}}(\mathbb{R}^k))$; define $(\alpha^*\omega)(\mathbf{x}) = T^*(\omega(p))$. ω is **of class** C^∞ **near** p if $\alpha^*\omega$ is of class C^∞ near $\mathbf{x}$ (independent of α), and of class C^∞ if this holds for every patch.

Henceforth all forms are C^∞. $\Omega^l(M)$ is the space of l-forms on M. There are no elementary forms on M giving a canonical form, but $\alpha^*\omega = \sum_{[I]} f_I\, dx_I$ in $\mathbb{R}^k$; the f_I are the **components** of ω with respect to α.

Definition 41.6 (differential)

For an l-form ω on M, $d\omega$ is defined as follows: given $p \in M$ and tangent vectors $\mathbf{v}_1, \dots, \mathbf{v}_{l+1}$ at p, choose a patch α about p with $\alpha(\mathbf{x}) = p$ and set

$$d\omega(p)(\mathbf{v}_1, \dots, \mathbf{v}_{l+1}) = d(\alpha^*\omega)(\mathbf{x})((\mathbf{x}; \mathbf{v}_1(\alpha)), \dots, (\mathbf{x}; \mathbf{v}_{l+1}(\alpha))).$$

That is, pull ω back to $\alpha^*\omega$ in $\mathbb{R}^k$, pull the $\mathbf{v}_i$ back to tangent vectors in $\mathbb{R}^k$, and apply d in $\mathbb{R}^k$. Equivalently (with $\mathbf{a}_i = \mathbf{v}_i(\alpha)$), $d\omega(p)(\alpha_*(\mathbf{x}; \mathbf{a}_1), \dots, \alpha_*(\mathbf{x}; \mathbf{a}_{l+1})) = d(\alpha^*\omega)(\mathbf{x})((\mathbf{x}; \mathbf{a}_1), \dots, (\mathbf{x}; \mathbf{a}_{l+1}))$, i.e. $d\omega$ is the unique $(l+1)$-form on M such that

$$\alpha^*(d\omega) = d(\alpha^*\omega)$$

for every coordinate patch α on M. (The d on the right is the operator in $\mathbb{R}^k$.)

This is independent of the patch, and $d\omega$ is C^∞.

Theorem 41.1

Let M be a compact differentiable manifold. Given a covering of M by coordinate patches, there exist functions $\phi_i : M \to \mathbb{R}$ of class C^∞, for $i = 1, \dots, l$, such that:
(1) $\phi_i(p) \geq 0$ for each $p \in M$.
(2) For each i, the set Support ϕ_i is covered by one of the given coordinate patches.
(3) $\sum \phi_i(p) = 1$ for each $p \in M$.

Proof. Given $p \in M$, choose a patch $\alpha : U \to V$ of the covering about p, $\alpha(\mathbf{x}) = p$, and a non-negative C^∞ function $f : U \to \mathbb{R}$ whose support is compact and contained in U, with $f(\mathbf{x}) > 0$. Define $\psi_p : M \to \mathbb{R}$ by $\psi_p(y) = f(\alpha^{-1}(y))$ for $y \in V$ and $\psi_p(y) = 0$ otherwise. Since $f \circ \alpha^{-1}$ vanishes outside a compact subset of V, ψ_p is C^∞ on M. ψ_p is positive on an open set U_p about p. Cover M by finitely many U_p, for $p = p_1, \dots, p_l$, and set $\lambda = \sum_{j=1}^{l} \psi_{p_j}$ and $\phi_i = (1/\lambda)\psi_{p_i}$. ∎

Definition 41.7 (integral of a form)

Let M be a compact oriented differentiable k-manifold and ω a k-form on M. If Support ω lies in a single patch $\alpha : U \to V$ of the orientation, $\int_M \omega = \int_{\mathrm{Int}\, U} \alpha^*\omega$. In general, with $\phi_1, \dots, \phi_l$ as in Theorem 41.1,

$$\int_M \omega = \sum_{i=1}^{l} \Big[\int_M \phi_i \omega \Big].$$

The usual argument shows this is well-defined and linear.

Theorem 41.2 (Stokes' theorem)

Let M be a compact, oriented differentiable k-manifold. Let ω be a $k-1$ form on M. If ∂M is non-empty, give ∂M the induced orientation; then

$$\int_M d\omega = \int_{\partial M} \omega.$$

If ∂M is empty, then $\int_M d\omega = 0$.

Proof. The proof of Theorem 37.2 goes through verbatim: all computations were carried out within coordinate patches, so no changes are needed. The special conventions for $k = 1$ and ∂M a 0-manifold are handled as before (Theorem 37.4). ■

Chapter 8 also generalizes: the **deRham group** $H^k(M)$ is the quotient of the closed k-forms on M by the exact ones, and there are methods to compute it, including a general *Mayer–Vietoris theorem* relating the deRham groups of $M = U \cup V$, U, V and $U \cap V$ (see Bott–Tu). $H^k(M)$ is obviously a diffeomorphism invariant of M. Strikingly, it is also a *topological* invariant: if M and N are homeomorphic, $H^k(M)$ and $H^k(N)$ are linearly isomorphic. This follows from *deRham's theorem*: the algebra of closed forms modulo exact forms on M is isomorphic to an algebra defined in algebraic topology for any topological space, the "cohomology algebra of M with real coefficients".

41.4 Riemannian manifolds

Not everything generalizes so readily: the volume of M and $\int_M f\,dV$ do not make sense on an abstract differentiable manifold. By §36, the volume of a compact oriented k-manifold M in $\mathbb{R}^n$ is $\mathrm{v}(M) = \int_M \omega_V$, where the volume form ω_V has value 1 on *any orthonormal basis* of $\mathcal{T}_p(M)$ in its natural orientation. This uses the inner product that $\mathcal{T}_p(M) \subset \mathcal{T}_p(\mathbb{R}^n) = p \times \mathbb{R}^n$ inherits from the dot product of $\mathbb{R}^n$; an abstract $\mathcal{T}_p(M)$ has none, so "orthonormal" is meaningless. We need an inner product on each tangent space.

Definition 41.8 (Riemannian metric)

A **Riemannian metric** on a differentiable k-manifold M is an inner product $\langle \mathbf{v}, \mathbf{w} \rangle$ on each tangent space $\mathcal{T}_p(M)$, required to be of class C^∞ as a 2-tensor field on M. A **Riemannian manifold** is a differentiable manifold together with a Riemannian metric.

("Metric" here has nothing to do with "metric space".) Every differentiable manifold has a Riemannian metric (the proof uses a partition of unity), but it is not unique. A Riemannian metric gives a volume function $V(\mathbf{v}_1, \ldots, \mathbf{v}_k)$ on k-tuples in $\mathcal{T}_p(M)$ (as in the exercises of §21), so one defines, for continuous f supported in a single patch $\alpha : U \to V$,

$$\int_M f\,dV = \int_{\mathrm{Int}\,U} (f \circ \alpha)\, V(\alpha_*(\mathbf{x}; \mathbf{e}_1), \ldots, \alpha_*(\mathbf{x}; \mathbf{e}_k)),$$

in general via a partition of unity as in §25, and $\mathrm{v}(M) = \int_M dV$. If M is compact, oriented and Riemannian, $\int_M \omega = \int_M \lambda\,dV$ where $\lambda(p)$ is the value of $\omega(p)$ on an orthonormal k-tuple in $\mathcal{T}_p(M)$ belonging to its natural orientation; if $\lambda \equiv 1$, ω is the **volume form** ω_V of M, and $\mathrm{v}(M) = \int_M \omega_V$.

Many questions arise: the *length* of a smooth curve $\gamma : [a, b] \to M$ is $\int_{t=a}^{t=b} \|\gamma_*(t; \mathbf{e}_1)\|$, the integral of the norm of its velocity vector (using the inner product on $\mathcal{T}_p(M)$); one studies *geodesics* (curves of minimal length joining two points), *curvature*, and so on—the subject of *Riemannian geometry*.

What does not generalize

Most of this book generalizes to abstract differentiable or Riemannian manifolds. The interpretation of Stokes' theorem in terms of scalar and vector fields (§38) does not: the "translation functions" of §31 depend on forms defined in $\mathbb{R}^n$, grad and div apply only to fields in $\mathbb{R}^n$, curl only in $\mathbb{R}^3$, and even the "normal vector" to a manifold depends on the surrounding space. Manifolds, differential forms and Stokes' theorem have meaning outside euclidean space; classical vector analysis does not.